A | Biophysical Neurovascular Coupling (NVC) Cascade

1. Synaptic Firing
Pyramidal Neurons
Glutamate Release
Temporal: ~10 ms



2. Astrocytic Relay
Astrocytic Endfeet
Ca2+ Elevation
Vasodilator Synthesis

EETs / NO / PGE2

3. Vascular Dilation
Arteriolar Smooth Muscle
Delta[HbO] Up, Delta[HbR] Down
Canonical Delay: ~5.0 s



<--- Biophysical Hemodynamic Delay Axis (tau ~ 5.16-5.25 s, Mean 5.18 s) --->

B | Multimodal Dual-Stream Acquisition & MBLL Inversion

Scalp EEG Stream

28 Electrodes (10-20 Standard Layout)
Sampling: 100 Hz | Common Average Reference (CAR)
Bandpass: 0.5 - 45.0 Hz (Zero-Phase SOS)



Prefrontal fNIRS & MBLL Inversion

36 Optode Pairs (Dual-Wavelength: 760 nm & 850 nm)
MBLL: Extinction Inversion Matrix Condition Number kappa = 3.64 < 15.0
Bandpass: 0.01 - 0.10 Hz (Mayer Wave & Cardiac Rejection)



C | NVC-DelayFormer Cross-Attention & Multi-Task Engine

EEG Spatiotemporal Encoder

1D Conv -> BatchNorm -> GELU
Keys & Values: K, V in R^(T_eeg x D)

fNIRS Convolutional Encoder

1D Conv -> Spatial BatchNorm -> GELU
Queries: Q in R^(T_fnirs x D)

Physiology-Guided Causal Cross-Attention Core

Attention Logits $S_{i,j} = (Q_i * K_j^T) / \sqrt{d_k} + \gamma * P_{i,j} + M_{i,j}$

Gaussian Delay Prior: $P_{i,j} = - ((t_i - t_j) - \tau)^2 / (2 * \sigma^2)$
[Learned: tau -> 5.18 s (5.16-5.25 s), sigma -> 1.50 s, gamma = 0.25]

Upper-Triangular Causal Mask: $M_{i,j} = 0$ if $(t_i - t_j >= 0)$ else -inf

Total Multi-Task Objective: $L_{total} = L_{ce} + 0.08 * ||w_{diff}||^2 + L_{recon}$

Task 1: Cognitive Classification

Linear Head -> y in {MA, Rest}
Loss: Cross-Entropy L_{ce}
Accuracy: 65.52% +- 10.58% (N=29)

Task 2: Auxiliary Delta[HbO] Decoder

1D Conv Reconstructor -> Y_HbO
Loss: MSE L_{recon} (r_{HbO} = 0.77)
Orthogonal Gradients (cos theta = +0.0089)